

Running Apache Hadoop over Lustre

A combination of the power of Hadoop and the speed of Lustre bodes well for enterprises. In this article, the author shows how Hadoop can be set up over Lustre. So those who love to play around with software, here's your chance!



Hadoop is a large-scale, distributed, open source framework for the parallel processing of data on large clusters built out of commodity hardware. While it can be used on a single machine, its true power lies in its ability to scale to hundreds or thousands of computers, each with several processor cores. Hadoop is also designed to efficiently distribute large amounts of work across a set of machines.

Hadoop is built in two main parts—a special file system called Hadoop Distributed File System (HDFS) and the MapReduce Framework. HDFS is an optimised file system for distributed processing of very large data sets on commodity hardware. HDFS stores data on the compute nodes, providing very high aggregate bandwidth across the cluster. Hadoop implements a computational paradigm named MapReduce, by

which the application is divided into many small fragments of work, which can each be executed or re-executed on any node in the cluster. Both MapReduce and the HDFS are designed so that node failures are automatically handled by the framework.

Hadoop runs on the master-slave architecture. An HDFS cluster consists of a single Namenode, a master server that manages the file system's namespace and regulates access to files by clients. There are a number of DataNodes, usually one per node in a cluster. The DataNodes manage storage attached to the nodes that they run on. HDFS exposes a file system's namespace and allows user data to be stored in files. A file is split into one or more blocks and a set of blocks are stored in DataNodes. DataNodes serve read and write requests, and perform block creation, deletion and replication upon instruction from Namenode.